

# Amirhossein Sourì

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## Education

### Sharif University of Technology

*B.Sc. in Computer Science and Engineering*

2022 – Expected June 2027

Tehran, Iran

- GPA: **18.99/20**
- Ranked **14th among 145,000+ participants** in Iran's National University Entrance Examination.

### Shahid Beheshti High School (NODET)

*Diploma in Mathematics and Physics*

2019 – 2022

Tehran, Iran

Affiliated with the National Organization for Development of Exceptional Talents

## Research Interests

- Machine and deep learning
- Computer vision and multimodal learning
- Language and vision-language models
- ML Systems
- Generative models
- Trustworthy ML
- Reinforcement learning and multi-agent systems
- Robotics

## Research Experience

### RIML Lab — Visual Reasoning in Vision-Language Models

July 2026 – Present

*Research Assistant (PI: Dr. Mohammad Hossein Rohban), Sharif University of Technology*

- Developed the project's initial question-conditioned visual-grounding study using CLIP patch/text features, heatmap-to-box conversion, and downstream evaluation with Qwen2.5-VL-3B-Instruct.
- Investigating training and evaluation methods for evidence-sensitive reasoning in vision-language models.

### RIML Lab — Diffusion Model Unlearning

Dec 2025 – Mar 2026

*Research Assistant (PI: Dr. Mohammad Hossein Rohban), Sharif University of Technology*

- Investigated noise-robust concept unlearning for text-to-image diffusion models under adversarial and distributional variation.
- Contributed to adaptive noise sampling and experimental evaluation; this work resulted in a preprint currently under review at ICLR 2027.

## Preprint

### Weeding Out Bad Seeds: Initial-Noise-Robust Unlearning for Text-to-Image Diffusion Models

Arian Komaei Koma, Seyed Amir Kasaei, Aida Aryafar, Matin Ghiasi, Ali Aghayari, **Amirhossein Sourì**, Mohammad Mosayyebi, AmirMahdi Sadeghzadeh, and Mohammad Hossein Rohban. *Under review at ICLR 2027.*

## Teaching Experience

### Sharif University of Technology

*Teaching Assistant*

Oct 2023 – Present

Tehran, Iran

- **Artificial Intelligence** — Dr. Mahdieh Soleymani *Fall 2025*
- **Artificial Intelligence** — Mr. Samiei and Mr. Fereydooni *Spring 2025*
- **Machine Learning** — Dr. Abolfazl Motahari *Spring 2025*
- **Probability and Statistics** — Dr. Amir Najafi *Spring 2024*
- **Advanced Programming (Java)** — Dr. MohammadAmin Fazli *Spring 2024, Spring 2026*
- **Fundamentals of Programming (C)** — Dr. MohammadAmin Fazli *Fall 2023, Fall 2024, Fall 2025*

## Industry Experience

### Hamravesh

*Software Engineer*

Oct 2024 – Sept 2025

Tehran, Iran

- Developed and deployed a marketplace service for one-click cloud applications, spanning backend APIs, frontend interfaces, and Kubernetes integration.

Selected Projects

- Question-Conditioned Visual Grounding | [GitHub](#) July 2026
  - Trained an attention-based model on CLIP patch and question embeddings to localize answer-relevant image regions.
  - Converted predicted heatmaps into bounding boxes and evaluated their effect on downstream VQA with Qwen2.5-VL-3B-Instruct.
- Deep Learning Practical Assignments | [GitHub](#) Spring 2026
  - Completed 15 implementation-focused notebooks spanning neural networks and optimizers from scratch, CNNs/RNNs, LoRA/QLoRA adaptation, RAG, and language model inference.
  - Implemented and evaluated VAE, DDPM, VQ-VAE-2, SimCLR, CLIP analysis, and latent-diffusion image-editing pipelines using PyTorch.
- Modern Information Retrieval Project | [GitHub](#) Spring 2026
  - Built a Goodreads search engine with preprocessing, near-duplicate detection, inverted/tiered indexes, BM25 and language model ranking, and MAP/NDCG/MRR evaluation.
  - Developed a hybrid multimodal product-search pipeline combining dense multimodal embeddings, SPLADE++ sparse retrieval, cross-encoder reranking, and structured LLM responses.
- Machine Unlearning and Robustness Exercises | [GitHub](#) Dec 2025
  - Implemented class unlearning in a conditional VAE using Fisher information and evaluated recovery attacks based on label manipulation, latent interpolation, and condition optimization.
- Artificial Intelligence Practical Assignments | [GitHub](#) Spring 2024
  - Implemented classical search, constraint satisfaction, minimax with alpha–beta pruning, Bayesian-network inference, HMMs, Q-learning, SARSA, and value iteration.
- Machine Learning Models Collection | [GitHub](#) Sept 2023
  - Implemented, explained, and compared core supervised and unsupervised methods—including regression, SVMs, trees, ensembles, dimensionality reduction, clustering, and Gaussian mixtures—with scikit-learn baselines.

Technical Skills

|                    |                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------|
| Programming        | Python, C/C++, Java, TypeScript/JavaScript, SQL, R                                                      |
| Machine Learning   | PyTorch, Transformers, Diffusers, PEFT, TRL, Accelerate, scikit-learn, SentenceTransformers, TensorFlow |
| Data & Vision      | NumPy, pandas, Hugging Face Datasets, FAISS, OpenCV, Matplotlib, Jupyter                                |
| Systems & Software | Linux, Git, Docker, Kubernetes, vLLM, Django/DRF, React, PostgreSQL                                     |
| Documentation      | LaTeX, Markdown                                                                                         |

Selected Coursework

|                              |                             |             |
|------------------------------|-----------------------------|-------------|
| Deep Learning                | Dr. Mahdieh Soleymani       | 19.7/20     |
| Machine Learning             | Dr. Ali Sharifi Zarchi      | 20.0/20     |
| Artificial Intelligence      | Dr. Mohammad Hossein Rohban | 19.3/20     |
| Modern Information Retrieval | Dr. Mahdieh Soleymani       | 20.0/20     |
| Computer Simulation          | Dr. Bardia Safaei           | In progress |
| Linear Algebra               | Dr. Maryam Ramezani         | 20.0/20     |
| Probability and Statistics   | Dr. Amir Najafi             | 19.2/20     |

Languages

Persian: Native proficiency      English: Professional working proficiency

Selected Academic Service & Leadership

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|-----------------------------------------------------------|---------------------|
| Public Relations Officer, CE Students’ Scientific Chapter | July 2025 – Present |
| Senior Editor, Byte Publication                           | May 2025 – Present  |
| Vice President & Technical Lead, Hardwar Event            | Feb 2025 – May 2025 |